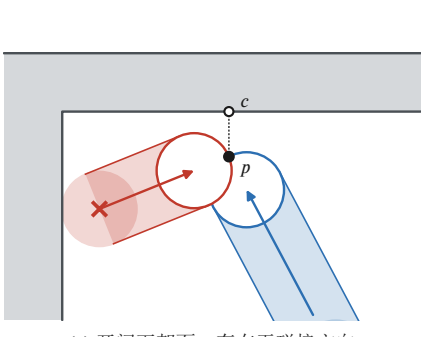
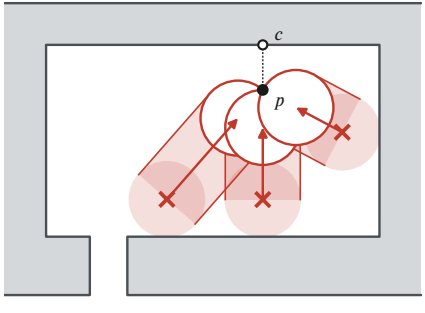
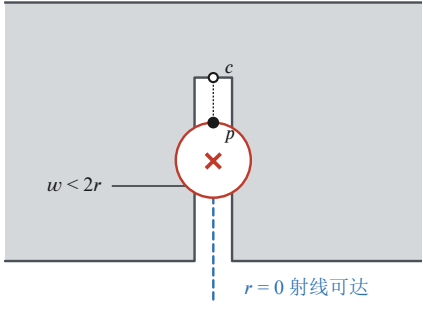


(a) 开阔下朝面：存在无碰撞方向



(b) 窄颈内腔：容得下工具球，但各方向均碰撞



(c) 窄缝：射线可见，有限半径工具不可达

零件实体　无碰撞扫掠　碰撞扫掠（× 为首次接触）　接触点 c　外移查询点 p